REPUBLIQUE DU CAMEROUN
PAIX – TRAVAIL – PATRIE
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MINISTERE DES ENSEIGNEMENTS SECONDAIRES
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REGION DE CENTRE

REPUBLICOF CAMEROON
PEACE – WORK – FATHERLAND
……..
MINISTRY OF SECONDARY EDUCATION
……..
CENTRE REGION

CASPA-INTER-REGIONAL MOCK EXAMINATION-CASPA-IRMEX BOARD	SUBJECT CODE NUMBER 0710	PAPER NUMBER 2
CANDIDATE NAME: …… CANDIDATE NUMBER: ……... CENTRE NUMBER: ……..	SUBJECT TITLE BIOLOGY	
ADVANCED LEVEL	DATE: March 26th 2024	

Time Allowed: 3h

Answer 5 questions out of the 8

All questions carry equal marks

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1. a) In maize, the genes for coloured seed and full seeds are dominant over the genes for colourless seeds and shrunken seeds. Pure breeding strains of the double dominant variety were crossed with the double recessive variety and a test cross on the F_1 generation produced the following.

- 380 coloured, full seeds
- 396 colourless , shrunken seeds
- 14 coloured , shrunken seeds
- 10 colourless, full seeds.

Calculate the distance in units between the genes for coloured seeds and seed shape.
b) Describe the process of selection in biology. **(5, 15 marks)**

2. a) state and explain the main functions of the kidney
b) Explain the general concept of the mechanism of homeostasis in mammals
c) Explain the events that occur during
i) Hypoglycemia
ii) Hyperglycemia **(5, 5, 10 marks)**

3. a) Define the following terms as used in reproduction
I. Fertilization
II. Pollination
III. Parthenogenesis
IV. Capacitation
V. Acrosome reaction
b) Describe the process in which sperm cells are produced in man. (**10, 10 marks)**

5. a) Describe in details the chemical structure of
I. A cocoyam tuber
II. Cow meat
b) Describe
I. The concept of isomerism in hexoses with structural illustrations.
II. Bonding in proteins (**10, 10 marks)**

6. Draw clearly annotated diagrams of
a) A longitudinal section through the human heart
b) The human alimentary canal
c) Section through the human eye (**6, 7, 7 marks)**

7. a) Describe the structure of a DNA molecule
b) Explain the concept of enzyme inhibition in metabolic reactions
c) Describe the process of glycolysis in tissue respiration. (**7, 5, 8 marks)**

8. a) Write a definition paragraph on each of the five major theories of the evolution of life on earth.
b) Write short notes on
I. Biotechnology
II. Genetic engineering (**10, 10 marks)**

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